CS 465 - Assignment 3

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1.

L_3

⟨a,b,c⟩
⟨a,b,d⟩
⟨a,c,d⟩
⟨b,c,d⟩
⟨b,c,w⟩
⟨b,c,x⟩
⟨p,q,r⟩
⟨p,q,s⟩
⟨p,q,t⟩
⟨p,r,s⟩
⟨q,r,s⟩

⇒

C_4^*

⟨a,b,c,d⟩ ← abc, abd, acd, bcd
⟨b,c,d,w⟩ ← bcd, bcw, bdw, cdw
⟨b,c,d,x⟩ ← bcd, bcx, bdx, cdx
⟨b,c,w,x⟩ ← bcw, bcx, bwx, cwx
⟨p,q,r,s⟩ ← pqr, pqs, prs, qrs
⟨p,q,r,t⟩ ← pqr, pqt, prt, qrt
⟨p,q,s,t⟩ ← pqs, pqt, pst, qst

C_4^{**}

⟨a,b,c,d⟩

⟨p,q,r,s⟩

C_4^{**} explanation: removed itemsets are marked in red.

	⟨b,c,d,w⟩	⟨b,c,d,x⟩	⟨b,c,w,x⟩	⟨p,q,r,t⟩	⟨p,q,s,t⟩
infrequent 3-item sets	b,d,w	b,d,x	b,w,x	p,r,t	p,s,t
	c,d,w	c,d,x	c,d,x	q,r,t	q,s,t

2.

$$\text{Supp}(X \to Y) = P(X \cup Y) \times 100 \quad , \quad \text{conf}(X \to Y) = \frac{P(X \cup Y)}{P(X)} \times 100$$

where

$$P(X \cup Y) = \frac{\text{count}(X \cup Y)}{N} \quad \& \quad P(X \cup Y) = P(Y \cup X) \quad , \quad P(X) = \frac{\text{count}(X)}{N}$$

thus

$$P(L \cup R) = \frac{\text{count}(L \cup R)}{5000} = \frac{3000}{5000} = 0.6 \Rightarrow \text{Supp}(L \to R) = \text{Supp}(R \to L) = P(L \cup R) \times 100 = .6 \times 100 = 60$$

$$P(L) = \tfrac{3400}{5000} = 0.68 \quad , \quad P(R) = \tfrac{4000}{5000} = 0.80 \Rightarrow \text{conf}(L \to R) = \frac{P(L \cup R)}{P(L)} \times 100 = \frac{0.6}{0.68} \times 100 \approx 88$$

$$\Rightarrow \text{conf}(R \to L) = \frac{P(L \cup R)}{P(R)} \times 100 = \frac{0.6}{0.8} \times 100 = 75$$

$\therefore$ $\text{Supp}(L \to R) = \text{Supp}(R \to L) = 60$

$\text{conf}(L \to R) \approx 88$

$\text{conf}(R \to L) = 75$